

Assignment 3 [written]

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1 Generative Adversarial Networks

(a) Vanilla GAN



Figure 1: Samples (GAN)



Figure 2: Interpolations (GAN)

Final FID (Full 50K): 163.37599389886697

Figure 3: Final FID (GAN)

- For GAN, even I finished the code under the guidance of the problem requirements, then after 25,000 iterations, FID score suddenly increased and collapsed as following plot, showing Gradient vanishing of Vanilla model. That caused the final FID will be more than 110.

- How to fix this unexpected phenomenon:

1. Adjust hyperparameters—lower learning rate/optimizer Adam/higher batch size

2. Different loss functions—actually what we did later

3. Limitation of D—Generator:Discriminator = 1:1/early stop D

So if I stop Discriminator at 26000 iterations, the final FID will be $100.36733 < 110$

Otherwise I need to change both lr and optimizer cause it was too high.



Figure 4: FID vs. Iterations (GAN)

(b) LS-GAN



Figure 5: Samples (LS)

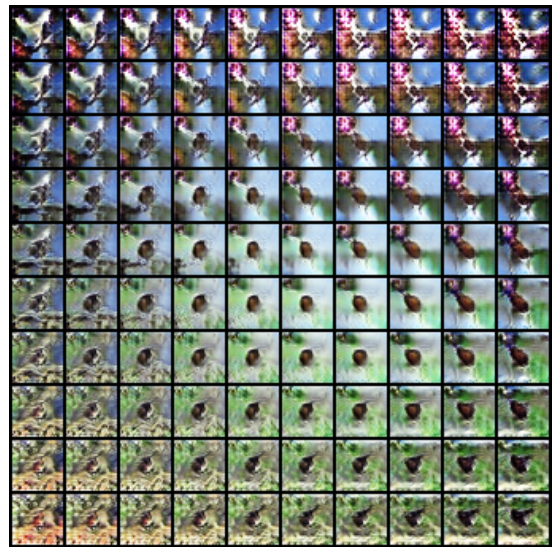


Figure 6: Interpolations (LS)

Final FID (Full 50K): 81.94507535983769

Figure 7: Final FID (LS)

(c) WGAN-GP



Figure 8: Samples (WGAN)

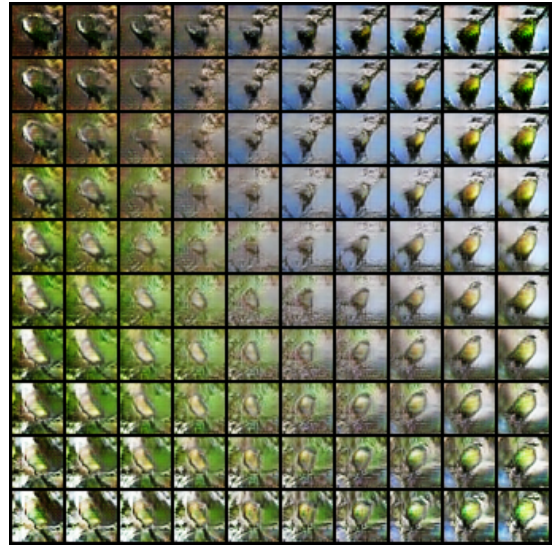


Figure 9: Interpolations (WGAN)

Final FID (Full 50K): 48.16291667320013

Figure 10: Final FID (WGAN)

2 Variational Autoencoders

(a) Autoencoder

- For size 16:

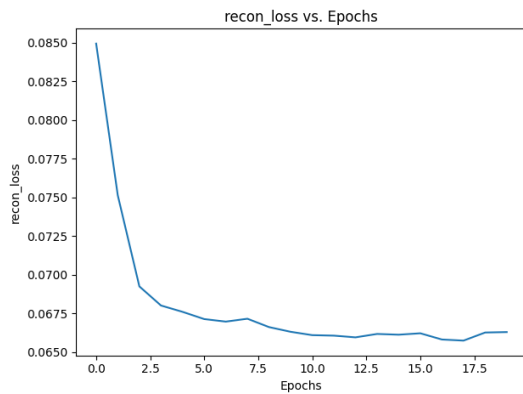


Figure 11: Loss vs. Iterations (Size 16)

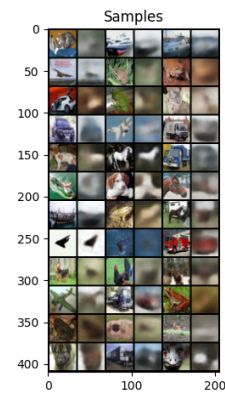


Figure 12: Reconstructions (Size 16)

- For size 128:

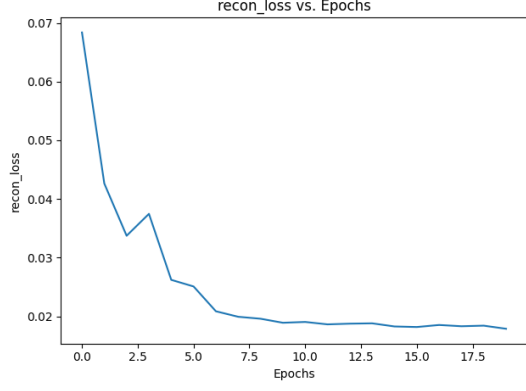


Figure 13: Loss vs. Iterations (Size 128)

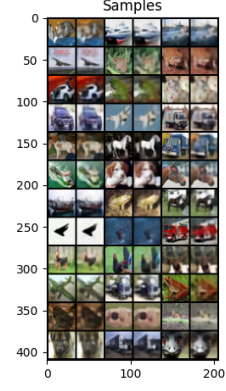


Figure 14: Reconstructions (Size 128)

- For size 1024:

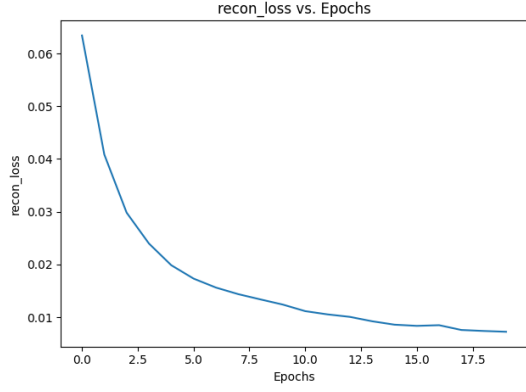


Figure 15: Loss vs. Iterations (Size 1024)

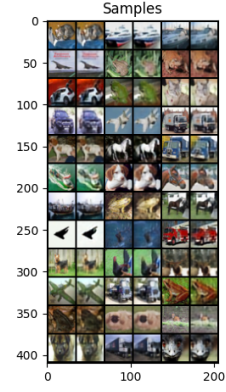


Figure 16: Reconstructions (Size 1024)

(b) VAE: After observing the results,

- $\beta^* = 0.8$: A lower weight for the KL divergence loss leads to the smallest reconstruction loss and the lowest sample blurriness.
- If $\beta = 1.2$: An excessively high weight for the KL divergence loss results in increased reconstruction loss and blurrier samples.
- If $\beta = 1.0$ (baseline): Reconstruction quality is less sharp than $\beta = 0.8$.

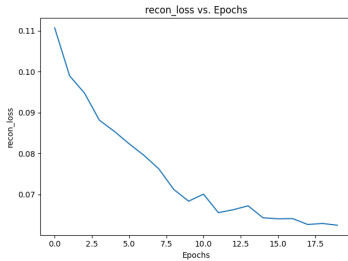


Figure 17: Recon.Loss (β^*)

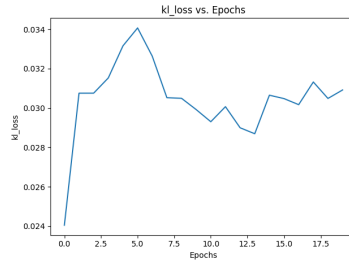


Figure 18: KL.Loss (β^*)



Figure 19: Sample (β^*)

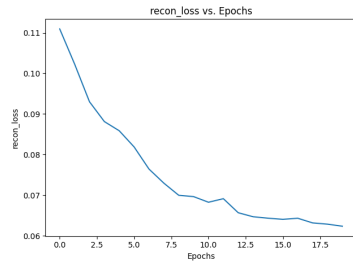


Figure 20: Recon.Loss (annealing)

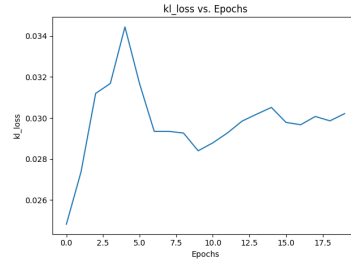


Figure 21: KL.Loss (annealing)



Figure 22: Sample (annealing)